

1. 设随机向量  $(X, Y)$  的联合概率密度为

$$f_{X,Y}(x, y) = \begin{cases} c(x^2 + xy + y^2), & 0 < x < 1, 0 < y < 1 - x, \\ 0, & \text{其他.} \end{cases}$$

(a) 求归一化常数  $c$ , 并显式写出边缘密度  $f_X(x)$ 、 $f_Y(y)$ 。

(b) 检验  $X$  与  $Y$  是否独立; 若不独立, 说明理由并给出一个数值代入的验证示例。

(c) 进行变量变换

$$U = X + Y, \quad V = \frac{X}{X + Y}$$

求联合密度  $f_{U,V}(u, v)$ 。

$$\begin{aligned} (a) \quad & \int_0^1 \left( \int_0^{1-x} c(x^2 + xy + y^2) dy \right) dx \\ &= \int_0^1 \left[ cx^2(1-x) + \frac{1}{2}cx(1-x)^2 + \frac{1}{3}c(1-x)^3 \right] dx \\ &= c \int_0^1 \left( x^2 - x^3 + \frac{1}{2}x - x^2 + \frac{1}{2}x^3 + \frac{1}{3} - x + x^2 - \frac{1}{3}x^3 \right) dx \\ &= c \int_0^1 \left( -\frac{5}{6}x^3 + x^2 - \frac{1}{2}x + \frac{1}{3} \right) dx \\ &= c \left( -\frac{5}{24} + \frac{1}{3} - \frac{1}{4} + \frac{1}{3} \right) \\ &= \frac{5}{24}c = 1 \\ &C = \frac{24}{5} \end{aligned}$$

$$\begin{aligned} f_X(x) &= \int_0^{1-x} c(x^2 + xy + y^2) dy \\ &= c(1-x) \left( x^2 + \frac{1}{2}x - \frac{1}{2}x^2 + \frac{1}{3} - \frac{2}{3}x + \frac{1}{3}x^2 \right) \\ &= \frac{24}{5}(1-x) \left( \frac{5}{6}x^2 - \frac{1}{6}x + \frac{1}{3} \right) = 4x^3 + \frac{24}{5}x^2 - \frac{12}{5}x + \frac{8}{5} \quad x \in (0, 1) \end{aligned}$$

$$f_Y(y) = \int_0^{1-y} c(x^2 + xy + y^2) dx$$

$$= -4y^3 + \frac{24}{5}y^2 - \frac{12}{5}y + \frac{8}{5} \quad y \in (0,1)$$

$$(b) \quad f_X(x)f_Y(y) \neq f_{X,Y}(x,y)$$

$$f(y|x=0) = \frac{f_{X,Y}(x=0,y)}{f_X(x=0)} = \frac{5}{8} \cdot \frac{24}{5}(y^2) = 3y^2 \neq f_{X,Y}$$

$$(c) \quad x = uv \quad y = u - uv \quad \left| \frac{d(x,y)}{d(u,v)} \right| = \begin{vmatrix} v & u \\ 1-v & -u \end{vmatrix} = u$$

$$f_{U,V}(u,v) = f_{X,Y}(x,y) \left| \frac{d(x,y)}{d(u,v)} \right|$$

$$= c[u^2v^2 + u^2v(1-v) + u^2(1-v)^2] u$$

$$= \frac{24}{5} [u^3v^2 + u^3v(1-v) + u^3(1-v)^2]$$

$$= \frac{24}{5} (u^3v^2 - u^3v + u^3) \quad u \in (0,1) \quad v \in (0,1)$$

$$2. (a) E(Y) = E\left(E(Y|X)\right)$$

$$= E_X(X)$$

$$= \frac{1}{2}$$

$$\text{Var} Y = E_X(\text{Var}_{Y|X}(Y|X)) + \text{Var}_X(E_{Y|X}(Y|X))$$

$$= E_X(X^2) + \text{Var}_X(X)$$

$$= \text{Var}(X) + (E(X))^2 + \text{Var} X$$

$$= 2 \times \frac{1}{12} + \frac{1}{4} = \frac{10}{24}$$

$$\text{COV}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E\left[\left(X - \frac{1}{2}\right)\left(Y - \frac{1}{2}\right)\right]$$

$$= E(XY) - \frac{1}{2}[E(X) + E(Y)] + \frac{1}{4}$$

$$E(XY) = E_X(E(XY|X))$$

$$= E(X^2)$$

$$= (E(X))^2 + \text{Var} X$$

$$= \frac{1}{3}$$

$$\therefore \text{COV}(X, Y) = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}$$

(b) 给定  $X \sim N(0, 1)$ ,  $Y \sim N(X, X^2)$   $\frac{Y}{X} \sim N(1, 1)$

$\therefore \frac{Y}{X} \sim N(1, 1)$   $X \sim N(0, 1)$ ,  $\frac{Y}{X}$  与  $X$  独立.